Dietary intake and risk of developing inflammatory bowel disease: a systematic review of the literature.

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OBJECTIVES: The incidence of inflammatory bowel disease (IBD) is increasing. Dietary factors such as the spread of the "Western" diet, high in fat and protein but low in fruits and vegetables, may be associated with the increase. Although many studies have evaluated the association between diet and IBD risk, there has been no systematic review. METHODS: We performed a systematic review using guideline-recommended methodology to evaluate the association between pre-illness intake of nutrients (fats, carbohydrates, protein) and food groups (fruits, vegetables, meats) and the risk of subsequent IBD diagnosis. Eligible studies were identified via structured keyword searches in PubMed and Google Scholar and manual searches. RESULTS: Nineteen studies were included, encompassing 2,609 IBD patients (1,269 Crohn's disease (CD) and 1,340 ulcerative colitis (UC) patients) and over 4,000 controls. Studies reported a positive association between high intake of saturated fats, monounsaturated fatty acids, total polyunsaturated fatty acids (PUFAs), total omega-3 fatty acids, omega-6 fatty acids, mono- and disaccharides, and meat and increased subsequent CD risk. Studies reported a negative association between dietary fiber and fruits and subsequent CD risk. High intakes of total fats, total PUFAs, omega-6 fatty acids, and meat were associated with an increased risk of UC. High vegetable intake was associated with a decreased risk of UC. CONCLUSIONS: High dietary intakes of total fats, PUFAs, omega-6 fatty acids, and meat were associated with an increased risk of CD and UC. High fiber and fruit intakes were associated with decreased CD risk, and high vegetable intake was associated with decreased UC risk.